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$x^2 - dy^2 = 1$ (positive Pell's equation)

$x^2 - dy^2 = -1$ (negative Pell's equation)

Units $\varepsilon = x + y\sqrt{d}$ in $\mathbb{Z}[\sqrt{d}]$ $\rightleftharpoons$ Solutions (x, y) to Pell's equations $x^2 - dy^2 = \pm 1$.

Thm. The units in $\mathbb{Z}[\sqrt{2}]$ are

$$\pm \varepsilon_0^n = \pm (1+\sqrt{2})^n, \quad n \in \mathbb{Z}.$$

Equivalently, the set of units under multiplication is an abelian group generated by -1 and ε_0.

Note: $\underbrace{(1+\sqrt{2})}_{\varepsilon_0}\underbrace{(-1+\sqrt{2})}_{\varepsilon_0^{-1}} = -1^2+(\sqrt{2})^2 = 1$

So $\varepsilon_0^{-1} = \dfrac{1}{1+\sqrt{2}} = \sqrt{2}-1 \in \mathbb{Z}[\sqrt{2}]$

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$\dfrac{-1+\sqrt{2}}{\underbrace{(1+\sqrt{2})(-1+\sqrt{2})}_{1}} = -1+\sqrt{2}$

$\varepsilon_0 = 1+\sqrt{2}$ satisfies $N(\alpha) = 1^2 - 2\cdot 1^2 = -1$

$N(\varepsilon_0^n) = (N(\varepsilon_0))^n = (-1)^n$, $n \geq 0$.

$N(\varepsilon_0^{-n}) = (N(\varepsilon_0^{-1}))^n = (N(-1+\sqrt{2}))^n = (-1)^n$, $n \geq 0$.

So $\pm \varepsilon_0^n$ $(n \in \mathbb{Z})$ are units in $\mathbb{Z}[\sqrt{2}]$.

Pf. Already verified that $\pm \varepsilon_0^n \ (n \in \mathbb{Z})$ are units.

Step 1. Show ε_0 is the least unit > 1.

Assume $1 < \varepsilon = x + y\sqrt{2} < \varepsilon_0 = 1 + \sqrt{2}$ is a unit. ①

So

$$|N(\varepsilon)| = |x^2 - 2y^2| = 1.$$

$\varepsilon \bar{\varepsilon} = (x + y\sqrt{2})(x - y\sqrt{2})$

Thus

$$|x - y\sqrt{2}| = \frac{1}{|x + y\sqrt{2}|} < 1. \Rightarrow -1 < x - y\sqrt{2} < 1. \quad ②$$

$|x + y\sqrt{2}| > 1$

① + ② : $0 < 2x < 2 + \sqrt{2} \Rightarrow 0 < x < 1 + \frac{\sqrt{2}}{2} < 2.$

So $x = 1$. But $x^2 - 2y^2 = \pm 1$. So $y = 0, \pm 1$.

Since $1 < \varepsilon = x + y\sqrt{2} < 1 + \sqrt{2}$, impossible.

Hence no unit lies in $(1, \varepsilon_0)$.

Step 2. Show that any unit is one of $\pm \varepsilon_0^n$ $(n \in \mathbb{Z})$.

Let ε be a unit. If $\varepsilon < 0$, we argue for $-\varepsilon > 0$.

If $0 < \varepsilon < 1$, then we argue for the unit $\varepsilon^{-1} > 1$.

So may assume $\varepsilon \geq 1$.



$\exists\, n \geq 0$ s.t. $\varepsilon_0^n \leq \varepsilon < \varepsilon_0^{n+1}$.

Then the unit $\varepsilon \varepsilon_0^{-n}$ lies in $[1, \varepsilon_0)$.

Step 1 shows that there is no unit in $(1, \varepsilon_0)$.

So $\varepsilon \varepsilon_0^{-n} = 1$, i.e., $\varepsilon = \varepsilon_0^n$. □

If $\varepsilon_n = \pm \varepsilon_0^n = x_n + y_n\sqrt{2}$, where $x_n, y_n \in \mathbb{Z}$, then

$$x_n^2 - 2y_n^2 = N(\varepsilon_n) = (N(\varepsilon_0))^n = (-1)^n.$$

So even values of n gives all $x, y \in \mathbb{Z}$ satisfying

$$x^2 - 2y^2 = 1,$$

and odd values of n gives all $x, y \in \mathbb{Z}$ satisfying

$$x^2 - 2y^2 = -1.$$

Thm (Primes in $\mathbb{Z}[\sqrt{2}]$) Let $\pi \in \mathbb{Z}[\sqrt{2}]$ be prime.

1. $\pi \mid 2$ with $|N(\pi)| = 2$! $\pm \varepsilon_0^n \sqrt{2}$ $(n \in \mathbb{Z})$.

2. $\pi = p \equiv \pm 3 \pmod 8$ or one of $\pm \varepsilon_0^n p$. $(n \in \mathbb{Z})$.

3. $\pi = x + y\sqrt{2} \mid p$ with $x, y \neq 0$, where $p \equiv \pm 1 \pmod 8$.

In fact,

$$|N(\pi)| = |\pi \overline{\pi}| = |x^2 - 2y^2| = p.$$

If π is such a prime divisor of p, there are ∞ prime divisors of p in total:

$$\pm \varepsilon_0^n \pi, \quad \pm \varepsilon_0^n \overline{\pi}.$$

So there are exactly ∞ ways of writing $p = |x^2 - 2y^2|$.

Ex. Find all units and classify all the primes in $\mathbb{Z}[\sqrt{3}]$.

Note: $x^2 - 3y^2 = -1$ has no solutions.

1. $x^2 + 1 = 3y^2$: $e_3(3y^2) = 1 + e_3(y^2) = 1 + 2e_3(y)$ is odd, so $3y^2 \neq \square + \square$, but $x^2 + 1$ is. So $x^2 - 3y^2 = -1$ has no integer solutions.

(Or just mod 3 : $x^2 \equiv -1 \pmod 3$, impossible).

2. $x^2 - 3y^2 = 1$ has a solution $\varepsilon_0 = 2 + \sqrt{3}$.

Show ε_0 is the least unit > 1 and every unit is one of $\pm \varepsilon_0^n$ $(n \in \mathbb{Z})$.

More generally, $x^2 - dy^2 = -1$ may not have a solution, but $x^2 - dy^2 = 1$ always have a solution $\varepsilon_0 = x_0 + y_0\sqrt{d}$, the least unit > 1.

fundamental unit: every unit in $\mathbb{Z}[\sqrt{d}]$ is one of $\pm \varepsilon_0^n$, when $x^2 - dy^2 = -1$ has no solutions.

If $x^2 - dy^2 = -1$ has a solution $\varepsilon_0' > 1$, the least unit > 1, then every unit is one of $\pm (\varepsilon_0')^n$.